Beyond the Dailey-Townes model: chemical information from the electric field gradient

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DIRAC MEETING











Presentation Outline

1. Introduction
2. Theory
3. Results
4. Conclusions
5. Supplementary material

Introduction

Introduction

Schüler and Schmidt (1935)[1]: first demonstrated case of **hyperfine structure** due to the nuclear electric quadrupole moment in ^{151}Eu and ^{153}Eu.

Nuclei with spin I>1/2 possess an **electric quadrupole moment** which measures the deviation from the spherical symmetry of the charge distribution.

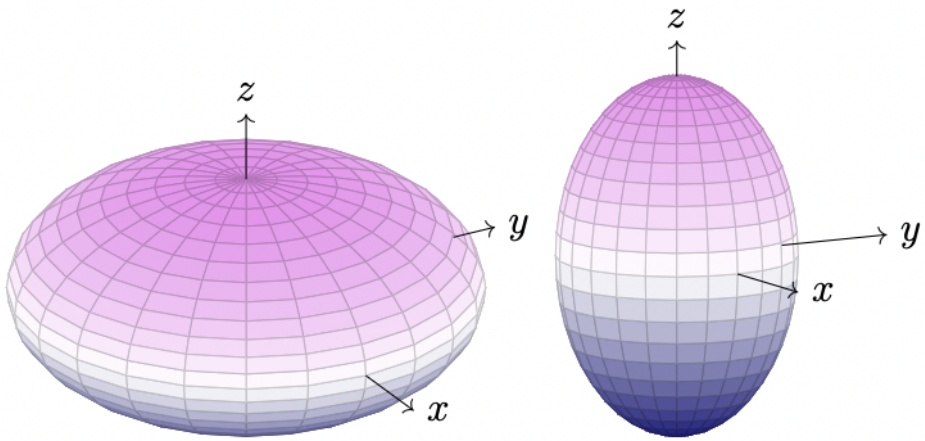


Figure 1: Nuclear electric quadrupole moment: oblate symmetry (left) and prolate symmetry (right).

[1] H. Schüler and T. Schmidt, Z. Phys, 94 (1935), 457–468.

Introduction

Introduction

The **nuclear quadrupolar coupling constant (NQCC)** defines the magnitude of this coupling[2]:

$$NQCC = \frac{e^2 qQ}{h} \tag{1}$$

- eQ is the zz component of nuclear electric quadrupole moment tensor;
- eq is the zz component of the electric field gradient tensor;

How we can determine the value of the nuclear electric quadrupole moment?

[2] **Pyykkö Pekka, Molecular Physics, 116 (2018), 1-11.**

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How we can determine the value of the nuclear electric quadrupole moment?

- **Experimentally** we can determine the value of NQCC:
  - Nuclear Magnetic Resonance, Microwave spectroscopy, Mössbauer spectroscopy, ...
- **Computationally** we can determine the value of eq:
  - Ab-initio approach, Density Functional Theory, ...

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Introduction

Goal

My work is based on the calculation of the **electric field gradient** at the nuclear positions in molecules, which allows us to obtain:

- **Chemical information**: electronic structure of chemical bonding;
- **Physical information**: nuclear quadrupole moment.

We want to compare our results in a relativistic framework with those obtained with the **Dailey-Townes model**[3].

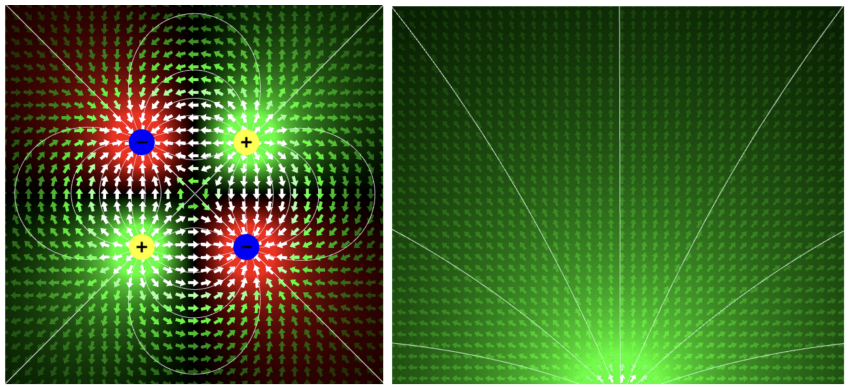


Figure 2: Nuclear electric quadrupole moment and electric field gradient.

[3] C.H. Townes and B.P. Dailey, J. Chem. Physics, 17(1949), 782-796.

Theory

Electric field gradient

The generic expression for one component of the **electric field gradient tensor** is:

$$\mathcal{E}^{[1]}_{\alpha\beta}(\boldsymbol{R}) = -\phi^{[2]}_{\alpha\beta}(\boldsymbol{R}) = -\frac{\partial^2 \phi(\boldsymbol{R})}{\partial r_\alpha \partial r_\beta} \tag{2}$$

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The electric field gradient at nucleus K is due to a **nuclear** and **electronic** contribution:

$$\langle \hat{\phi}^{[2]}_{\alpha\beta}(\boldsymbol{R}_K) \rangle = \underbrace{-\frac{e}{4\pi\epsilon_0} \sum_i^{N_{occ}} \left\langle \frac{3\hat{r}_{iK;\alpha}\hat{r}_{iK;\beta} - \delta_{\alpha\beta}\hat{r}^2_{iK}}{\hat{r}^5_{iK}} \right\rangle}_{\text{electronic}} + \underbrace{\sum_{A\neq K} \frac{Z_A e}{4\pi\epsilon_0} \left[\frac{3R_{K;\alpha}R_{K;\beta} - \delta_{\alpha\beta}R^2_K}{R^5_K} \right]}_{\text{nuclear}} \tag{3}$$

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The electronic contribution at the SCF level can be expressed as a sum of contributions deriving from all molecular orbitals:

$$\langle\hat{\phi}^{[2],el}_{\alpha\beta}\rangle = \sum_i^{Nocc}\langle\Psi^{MO}_i|\hat{\phi}^{[2],el}_{\alpha\beta}|\Psi^{MO}_i\rangle \tag{4}$$

One way to extract chemical information from the electric field gradient is to use the so called **Dailey-Townes model**.

Theory

Dailey-Townes model

The assumption for the Dailey-Townes model

$$\langle \psi_{n\ell m} | e\hat{q}_K^{el} | \psi_{n\ell m} \rangle = \frac{4}{5} e \langle r^{-3} \rangle \left[P_{zz} - \frac{P_{xx} + P_{yy}}{2} \right] \tag{5}$$

The **valence *p* orbitals that are not fully occupied** as the orbitals that contribute most to the electric field gradient; in our notation $e\hat{q}_K^{el} = \hat{\phi}_{zz}^{[2],el}(\boldsymbol{R}_K)$. $\boldsymbol{P}$ is the population density matrix that represent the population of the p atomic orbitals.

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1. s orbitals are **spherically symmetric** and therefore the electric field gradient is zero.

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Justifications

1. s orbitals are **spherically symmetric** and therefore the electric field gradient is zero.
2. From the **Unsöld's theorem** any complete atomic shell becomes spherically symmetric and therefore does not contribute to the field gradient:

$$\sum_{m=-\ell}^{+\ell} |Y_{\ell m}(\theta, \phi)|^2 = \frac{2\ell + 1}{4\pi} \Rightarrow \sum_{m=-\ell}^{+\ell} \langle Y_{\ell m} | Y_{20} | Y_{\ell m} \rangle = 0 \tag{6}$$

Theory

Projection analysis

In our work we used the projection analysis [4] to expand the molecular orbital:

$$|\Psi_i^{MO}\rangle = \sum_A \sum_{p \in A} c_{p,i}^A \, |\Psi_p^A\rangle + |\Psi_i^{pol}\rangle \tag{7}$$

[4] S. Dubillard, J. B. Rota, T. Saue and K. Faegri, . Chem. Physics, 124(2006)

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$$\begin{aligned} \langle e\hat{q}_K^{el}\rangle = \sum_i^{Nocc} \langle \Psi_i^{MO}|e\hat{q}_K^{el}|\Psi_i^{MO}\rangle &= \left(\langle e\hat{q}_K^{el}\rangle_{intra}^{(A)} + \langle e\hat{q}_K^{el}\rangle_{inter}^{(AB)} + \langle e\hat{q}_K^{el}\rangle_{pol}\right) \\ &= \left(\langle e\hat{q}_K^{el}\rangle_{princ}^{(A)} + \langle e\hat{q}_K^{el}\rangle_{hyb}^{(A)} + \langle e\hat{q}_K^{el}\rangle_{inter}^{(AB)} + \langle e\hat{q}_K^{el}\rangle_{pol}\right) \end{aligned} \tag{8}$$

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Results

Analysis of ICI

Results

Analysis of ICl: what the results say

Table 1: DIRAC output for the calculation of the electric field gradient for ICl. The electronic, nuclear and total contributions for the components of the electric field gradient tensor other than zero are reported. The expectation values are reported in a.u.

Nucleus ^{35}Cl	Electronic	Nuclear	Total
$\phi_{xx}^{[2]}$	-1.7409	-0.6273	-2.3682
$\phi_{yy}^{[2]}$	-1.7409	-0.6273	-2.3682
$\phi_{zz}^{[2]}$	3.4818	1.2546	4.7365

The electric field gradient must satisfy the following conditions:

$$\begin{cases} \textbf{Traceless} & \sum_i^3 \phi_{ii}^{[2]} = 0 \\ \textbf{Axial symmetry} & \phi_{xx}^{[2]} = \phi_{yy}^{[2]} \end{cases} \Rightarrow \phi_{xx}^{[2]} = \phi_{yy}^{[2]} = -\frac{1}{2}\phi_{zz}^{[2]}$$

Results

Analysis of ICl: what the results say

Table 2: DIRAC output for the calculation of the electronic part for the electric field gradient using projection analysis for ICl. The expectation values are reported in a.u.

Intratomic			
	Cl		
		Principal	**5.890**
		Hybridization	-0.570
	I		
		Principal	-1.212
		Hybridization	0.993×10^{-2}
Interatomic			
	I		-0.166×10^{-1}
Polarization			
	Total		-0.619

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$$\langle e\hat{q}_I^{el}\rangle_{prin} + e\hat{q}_I^{nucl} = -1.212 + 1.255 \simeq \mathbf{0} \quad \text{a.u}$$

Results

Analysis of ICl: what the results say

Table 3: Origin of the principal component of the electric field gradient at the position of the nucleus Cl. Each matrix element is multiplied by the density matrix. The results are reported in a.u.

Matrix element	Value	Sum
$\langle 1s \vert e\hat{q}^{el}_{Cl} \vert 1s \rangle$	0	0
$\langle 2s \vert e\hat{q}^{el}_{Cl} \vert 2s \rangle$	0	0
$\langle 2p_x \vert e\hat{q}^{el}_{Cl} \vert 2p_x \rangle$	81.586	
$\langle 2p_z \vert e\hat{q}^{el}_{Cl} \vert 2p_z \rangle$	-163.186	
$\langle 2p_y \vert e\hat{q}^{el}_{Cl} \vert 2p_y \rangle$	81.586	
		0
$\langle 3s \vert e\hat{q}^{el}_{Cl} \vert 3s \rangle$	0	0
$\langle 3p_x \vert e\hat{q}^{el}_{Cl} \vert 3p_x \rangle$	6.099	
$\langle 3p_z \vert e\hat{q}^{el}_{Cl} \vert 3p_z \rangle$	-6.294	
$\langle 3p_y \vert e\hat{q}^{el}_{Cl} \vert 3p_y \rangle$	6.099	
		5.912

Results

Analysis of chemical bond in $[O{\equiv}U{\equiv}O]^{2+}$

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Atom		Charge	Occupation of atomic orbitals
U	Ionic	+6	$[Rn]6p^6 5f^0 6d^0 7s^0$
	DFT results	+2,93	$[Rn]6p^{5.64} 5f^{2.29} 6d^{1.2} 7s^{0.003}$
O	Ionic	-2	$1s^2 2s^2 2p^6$
	DFT results	-0,31	$1s^2 2s^{1.9} 2p^{4.41}$

Table 4: Analysis of the electronic structure of UO_2^{2+} from the purely ionic point of view to that obtained from the results conducted with DFT.

- The calculation shows a **covalent** bond situation with respect to the ionic figure.
- **6*p*-hole**: overlap between U($6p$) with O($2s$) and O($2p$).

Results

Projection analysis

Intraatomic			
	U		
		Principal	-1.0654
		Hybridization	0.4860
	O		
		Principal	-0.4569
		Hybridization	0.0210
Interatomic			
	O		0.5901
Polarization			
	Total		**-7.2004**

- We have to explain the origin of the **high contribution of polarization**.

Results

An unexpected result

- Investigating the expectation value of each MOs we notice this:

```
          Expectation value for individual orbitals
          -----------------------------------------
               Matrix element    Occ.
E1   1   0.45959414E-01   2.0000   0.91918828E-01
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- We immediately notice an inconsistency: the LOMO contribution **is not zero**.
- Due to the **difference in energies**, we are expecting that the LOMO of the uranyl is a pure U(s):

$$\varepsilon_{1s}(\mathrm{U}) = -4256.3917 \quad \text{a.u}$$
$$\varepsilon_{1s}(\mathrm{O}) = -18.9019 \quad \text{a.u}$$

Results

The LOMO problem

- Well, let's try to see what happen with the simple **uranium atom**:

```
          Expectation value for individual orbitals
          -----------------------------------------
              Matrix element     Occ.
E1   1   -0.22335371E-06   2.0000   -0.44670742E-06
```

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```

- This suggests to us the **environment** around uranium in uranyl is very important:

$$\mathrm{O} \equiv \mathrm{U}^{2+} \equiv \mathrm{O}$$

- Due to the formation of the U-O bond we observe **a small deformation of U(1s)**.

Results

The polarization effect

	SREL		SOC	
AO	$\langle eq \rangle$	$\sum$	$\langle eq \rangle$	$\sum$
U 1s	0.108	0.108	0.108	0.108
U 2s	-0.006	0.102	-0.001	0.107
U 2p	-0.675	-0.573	-0.716	-0.609
U 3s	-0.219	-0.792	-0.281	-0.890
U 3p	-1.886	-2.678	-2.002	-2.892
U 3d	-0.070	-2.748	0.016	-2.876
U 4s	-0.482	-3.230	-0.588	-3.464
U 4p	-1.403	-4.633	-1.480	-4.944
U 4d	0.339	-4.294	0.464	-4.480
O 1s (2x)	-0.239	-4.533	-0.239	-4.719
U 4f	-0.238	-4.771	-0.237	-4.956
U 5s	-0.657	-5.428	-0.777	-5.733
U 5p	0.992	-4.436	0.916	-4.817
U 5d	1.066	-3.370	1.166	-3.651
U 6p / O 2s (2x)	13.912	**10.542**	13.674	**10.023**
O 2p (2x) / U 5f / U 6d	-17.806	**-7.264**	-18.25	**-8.227**

Results

Analysis of chemical bond in $[O{\equiv}U{\equiv}O]^{2+}$

UO_2^{2+} is a polyatomic molecule: to 'see' the chemical bonds it is necessary to **localized** the MOs:

	MO	AOs	G.c
Canonical	HOMO-11	U($6p$)	1.080
		O_1($2s$)	0.349
		O_2($2s$)	0.349
	HOMO-6	U($6p$)	0.531
		O_1($2s$)	0.593
		O_2($2s$)	0.593
Localized	HOMO-11	U($6p$)	0.757
		O_2($2s$)	0.113
	HOMO-6	O_1($2s$)	1.907

Table 5: Molecular orbitals in the UO_2^{2+} molecule involving the U($6p$) and O($2s$) orbitals. G.c=Gross contribution.

Results

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After localization **U(6*p*) hole tends to close**:

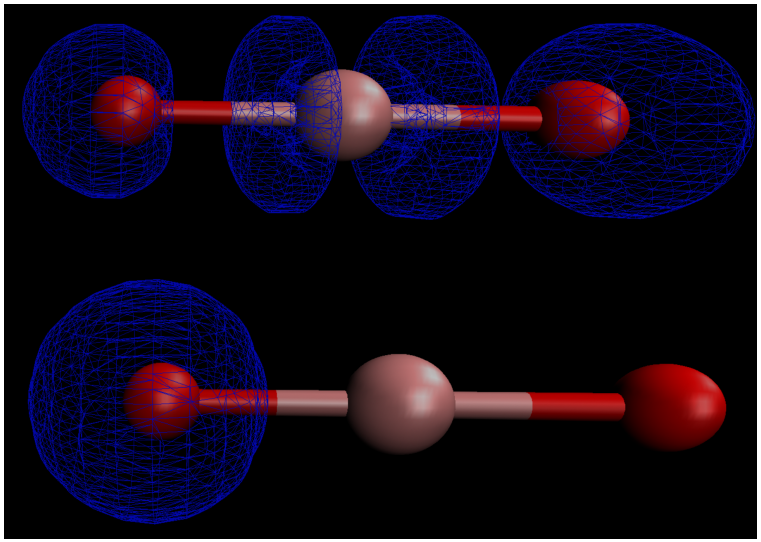


Figure 3: HOMO-6.

MOs	*eq* U(6*p*)-shell (a.u)
Canonical	≈ -82.870
Localized	≈ 1.742

Table 6: Total EFG contribution of AOs U(6*p*) from HOMO-11 to HOMO at ^{238}U.

Working on

Localization problem

- To localize the MOs we choose the **Pipek-Mezey criteria**: minimize the number of centers spanned by each molecular orbital.

$$G^{PM}\{\psi_i\} = -\sum_{i=1}^{occ}\sum_{A=1}^{nuclei}(Q_A^i)^2 \tag{9}$$

- If we perform this localization procedure on uranyl, we see the rotation of the core orbitals: **they are already strictly localized by definition!**

Conclusions

Conclusions

1. The electric field gradient turned out to be very sensitive to deviations from spherical symmetry, **such arising from bond formation**.
2. The results obtained for the ICl molecule are **consistent** with those of Dailey-Townes.
3. For UO_2^{2+}, in order to have an accurate description of the electric field gradient, we need also to take in to account ***d***, ***f*** and **core** orbitals.
4. The localization procedure tends **to close the U(6p)** orbital, confirmed by the low contribution to EFG.

Acknowledgements





HAMP-vQED team

X-ray guy

European Research Council
Established by the European Commission

Thank you for your attention

Questions?

Supplementary material

Dailey-Townes model

The electronic contribution to the *zz* component of the electric field gradient at the position of the nucleus K is:

$$\langle e\hat{q}_K^{el} \rangle = \sum_i^{Nocc} \left\langle \Psi_i^{MO} \left| \frac{-e}{4\pi\epsilon_0} \left(\frac{3z_{i;K}^2 - r_{i;K}^2}{r_{i;K}^5} \right) \right| \Psi_i^{MO} \right\rangle \tag{10}$$

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We consider the nucleus K at the origin:

$$\frac{3z^2 - r^2}{r^5} = \frac{3\cos^2\theta - 1}{r^3} = \sqrt{\frac{16\pi}{5}} \frac{Y_{20}(\theta, \phi)}{r^3} \tag{11}$$

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Non-relativistic atomic orbitals are introduced: $\Psi_{n\ell m}(\boldsymbol{r}) = R_{n\ell}(r)Y_{\ell m}(\theta,\phi)$:

$$\langle \Psi_{n_1\ell_1 m_1} | e\hat{q}_K^{el} | \Psi_{n_3\ell_3 m_3} \rangle = \frac{-e}{4\pi\epsilon_0} \sqrt{\frac{16\pi}{5}} \langle R_{n_1\ell_1} | r^{-3} | R_{n_3\ell_3} \rangle \langle Y_{\ell_1 m_1} | Y_{20} | Y_{\ell_3 m_3} \rangle \tag{12}$$

Supplementary material

Dailey-Townes model

Using the **Clebsch–Gordan coefficients** for the angular part:

$$\langle Y_{\ell_3 m_3} | Y_{\ell_2 m_2} | Y_{\ell_1 m_1} \rangle = \left[\frac{(2\ell_1 + 1)(2\ell_2 + 1)}{4\pi(2\ell_3 + 1)} \right]^{1/2} \langle \ell_1 m_1 \ell_2 m_2 | \ell_3 m_3 \rangle \langle \ell_1 0 \ell_2 0 | \ell_3 0 \rangle \qquad (13)$$

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In our case $\ell_2 = 2, m_2 = 0$ and we obtain two **selection rules**:

1. $m_3 = m_1$;
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1. $m_3 = m_1$;
2. $\ell_3 \in [\ell_1 - 2, ..., \ell_1 + 2]$.

Within the same atomic shell ($\ell_1 = \ell_3 = \ell$) only **diagonal** terms contribute.

$$\langle \Psi_{n\ell m} | e\hat{q}_K^{el} | \Psi_{n\ell m} \rangle = \frac{e}{2\pi\epsilon_0} \langle r^{-3} \rangle \frac{3m^2 - \ell(\ell + 1)}{(2\ell - 1)(2\ell + 3)} \quad (14)$$

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Dailey-Townes model

From the **Unsöld's theorem** any complete atomic shell becomes spherically symmetric and therefore does not contribute to the field gradient:

$$\sum_{m=-\ell}^{+\ell} |Y_{\ell m}(\theta,\phi)|^2 = \frac{2\ell+1}{4\pi} \Rightarrow \sum_{m=-\ell}^{+\ell} \langle Y_{\ell m}|Y_{20}|Y_{\ell m}\rangle = 0 \tag{15}$$

Supplementary material

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The assumption for the Dailey-Townes model

The **valence *p* orbitals that are not fully occupied** are the orbitals that contribute most to the electric field gradient.

$$\langle \psi_{n\ell m}|e\hat{q}_K^{el}|\psi_{n\ell m}\rangle = \frac{4}{5} e \langle r^{-3}\rangle \left[P_{zz} - \frac{P_{xx}+P_{yy}}{2}\right] \tag{16}$$

where the diagonal terms of the population density matrix $\mathcal{P}$ represent the population of the atomic orbitals.

Supplementary material

Spin-orbit coupling in ICl

Table 7: Origin of the principal component of the electric field gradient at the position of the nucleus Cl with spin-orbit coupling. Each matrix element is multiplied by the density matrix. The results are reported in a.u.

Matrix element	Value	Sum
$\langle 1s_{1/2,1/2}\|e\hat{q}^{el}_{Cl}\|1s_{1/2,1/2}\rangle$	0	0
$\langle 2s_{1/2,1/2}\|e\hat{q}^{el}_{Cl}\|2s_{1/2,1/2}\rangle$	0	0
$\langle 2p_{1/2,1/2}\|e\hat{q}^{el}_{Cl}\|2p_{1/2,1/2}\rangle$	0	
$\langle 2p_{3/2,1/2}\|e\hat{q}^{el}_{Cl}\|2p_{3/2,1/2}\rangle$	-81.00	
$\langle 2p_{3/2,-3/2}\|e\hat{q}^{el}_{Cl}\|2p_{3/2,-3/2}\rangle$	81.00	
		0
$\langle 3s_{1/2,1/2}\|e\hat{q}^{el}_{Cl}\|3s_{1/2,1/2}\rangle$	0	0
$\langle 3p_{1/2,1/2}\|e\hat{q}^{el}_{Cl}\|3p_{1/2,1/2}\rangle$	0	
$\langle 3p_{3/2,-3/2}\|e\hat{q}^{el}_{Cl}\|3p_{3/2,-3/2}\rangle$	6.0587	
$\langle 3p_{3/2,1/2}\|e\hat{q}^{el}_{Cl}\|3p_{3/2,1/2}\rangle$	-4.304	
		1.754

Supplementary material

Spin-orbit coupling in ICl

Table 8: Origin of the hybridization component of the electric field gradient at the position of the nucleus Cl with spin-orbit coupling. The results are reported in a.u.

Matrix element	Value	Sum
$\langle 3p_{3/2,1/2} \vert e\hat{q}^{el}_{Cl} \vert 3p_{1/2,1/2} \rangle$	4.304	
$\langle 3p_{1/2,1/2} \vert e\hat{q}^{el}_{Cl} \vert 2p_{1/2,1/2} \rangle$	-0.1946	
$\langle 3p_{3/2,1/2} \vert e\hat{q}^{el}_{Cl} \vert 2p_{1/2,1/2} \rangle$	-0.1911	
$\langle 3p_{3/2,3/2} \vert e\hat{q}^{el}_{Cl} \vert 2p_{3/2,1/2} \rangle$	-0.1341	
		3.491

Why with the spin-orbit coupling do we unexpectedly get a higher value of the hybridization contribution than the principal one?

Supplementary material

Spin-orbit coupling in ICl

When the atom is inserted in an molecule, is that spin-orbit coupling is so-called **quenched**, or reduced, by the molecular field.

- For the electric field gradient we observe that the **inner** molecular orbitals contribute very little since they form closed shells.
- Using Mulliken population analysis, we will see that these MOs generally do not have pure spin.
- However, the contributions to the electric field gradient tend to come from **valence orbitals**: these are almost **pure spin**, so an LCAO description is perhaps better done in spinfree AOs.

Supplementary material

The LOMO problem in UO_2^{2+}

- We have to discriminate the different contributions of the basis function based on the symmetry:

$$\langle LOMO|e\hat{q}|LOMO\rangle = \sum_{i,j}^{N_s} \langle U(s)_i|e\hat{q}|U(s)_j\rangle + \sum_{i,j}^{N_d} \langle U(d)_i|e\hat{q}|U(d)_j\rangle$$

$$2\sum_{i}^{N_s}\sum_{j}^{N_d} \langle U(s)_i|e\hat{q}|U(d)_j\rangle$$

Supplementary material

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$$\underbrace{2\sum_{i}^{N_s}\sum_{j}^{N_d}\langle U(s)_i|e\hat{q}|U(d)_j\rangle}_{\neq 0}$$

Supplementary material

The LOMO problem in UO_2^{2+}

- Analyzing the expectation value we can see

$$\langle LOMO|e\hat{q}|LOMO\rangle = 2\sum_{i}^{N_s}\sum_{j}^{N_d}\langle U(s)_i|e\hat{q}|U(d)_i\rangle = 0.045959414$$

Why the contribution $s - d$ is so big?

Supplementary material

The LOMO problem in UO_2^{2+}

- We are using **Gaussian-type orbitals**:

$$G_{ijk}^{\alpha}(\boldsymbol{r}) = N_{ijk}^{\alpha} x^i y^j z^k e^{-\alpha r^2} \;\; ; \;\; i+j+k = l$$

Supplementary material

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- For the uranium atom we have (dyall.v3z basis):

$$\alpha_s = 5.66886267 \cdot 10^7$$
$$\alpha_d = 1.10309490 \cdot 10^5$$

Supplementary material

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- Even if the deformation is small, the matrix elements involving U(s) and U(d) functions are large:

$$\langle U(s)|e\hat{q}|U(d)\rangle = 5.668 \cdot 10^7 \text{a.u}$$